BLOCKCHAIN

Blockchain Basics Visual Guide

Blocks, hashes, Merkle trees, and consensus — how trust works without a trusted party.

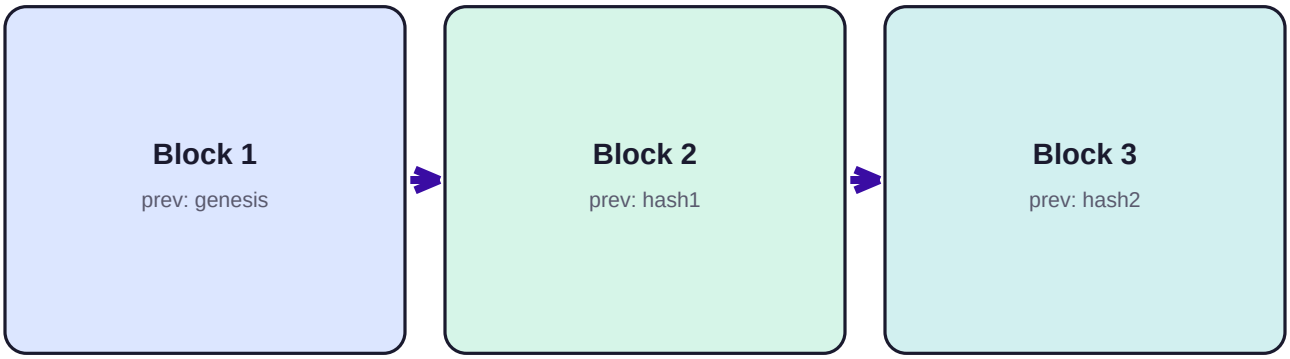
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Blocks & Hashing: The Chain

Each block commits to the previous block's hash — tampering breaks the link.



Change one byte in Block 1 -> its hash changes -> Block 2's 'prev' no longer matches.

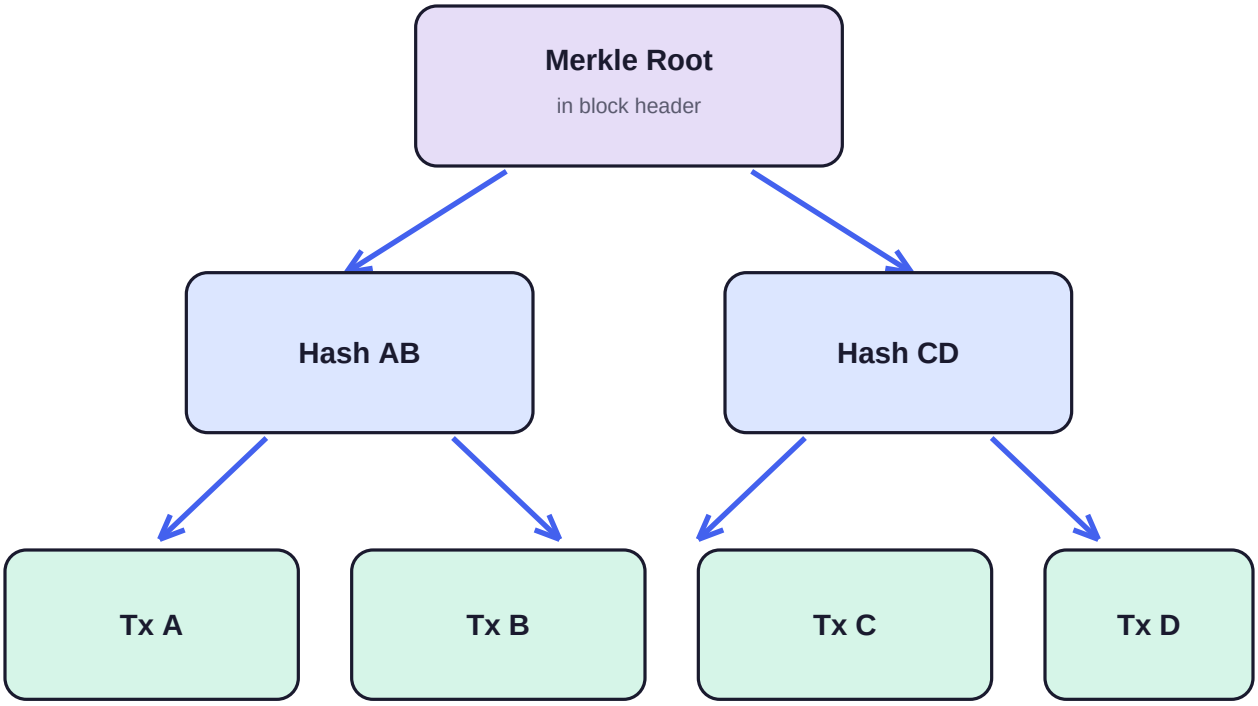
IMMUTABILITY

rewriting history means recomputing every later block — infeasible at scale

STEP 2/6

Merkle Trees

Hash transactions pairwise up to one root — prove inclusion with log(n) hashes.

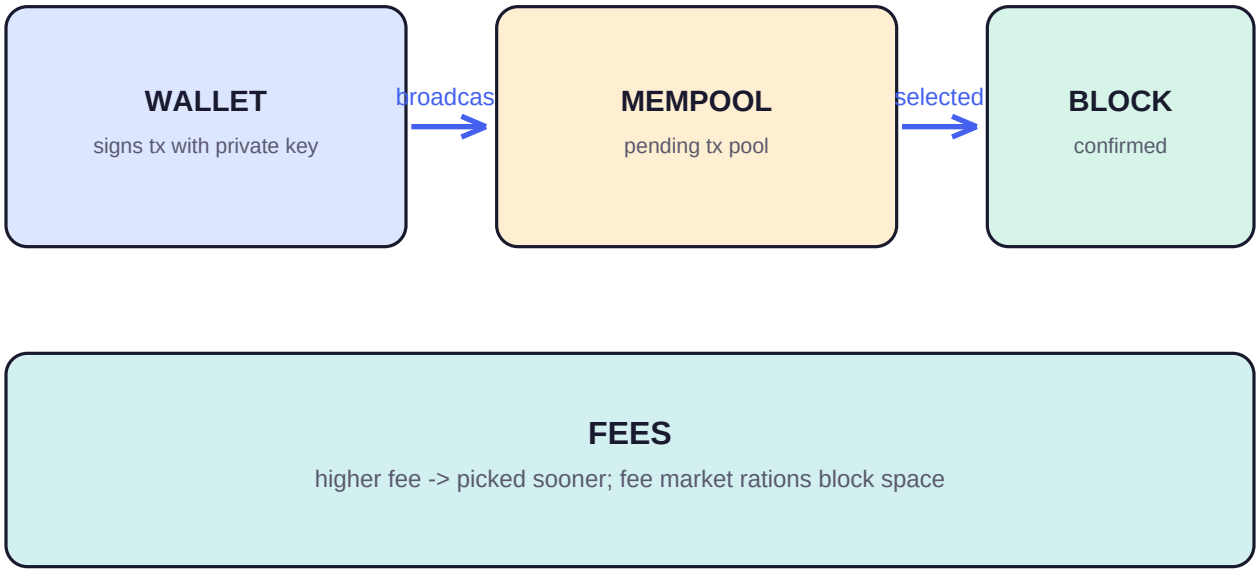


Light clients verify a transaction with just the root + log(n) sibling hashes — no full block needed.

STEP 3/6

Transactions & the Mempool

Signed intents wait in the mempool; miners/validators pick them into blocks.



Signatures prove authorization; the mempool is just a waiting room with an auction.

STEP 4/6

Consensus: PoW vs PoS

How strangers agree on one history without a referee.

PROOF OF WORK

burn electricity racing to find a valid hash

PROOF OF STAKE

lock capital; validators chosen by stake, slashed if dishonest

SYBIL RESISTANCE

both make fake identities expensive — via energy or via capital

51% ATTACK

control the majority of work/stake -> rewrite recent history

STEP 5/6

Finality: When Is It Really Done?

Probabilistic vs absolute — know which chain you are on.

PROBABILISTIC

PoW: deeper = safer; 6 confirmations ~ settled

ABSOLUTE

BFT PoS: 2/3 vote -> finalized, never reverted

Forks happen: two valid blocks at once. The fork-choice rule picks the canonical chain.

Exchanges wait N confirmations because finality is a confidence level, not a switch — on PoW chains.

STEP 6/6

Nodes & the Network

No center: full nodes verify everything, light nodes verify proofs.

Full nodes	store + validate every block — the network's immune system
Light clients	headers + Merkle proofs — phones and browsers
Miners / validators	produce blocks under the consensus rules
Gossip network	blocks and txs flood peer-to-peer; no central relay

STEP ✓I✓

Key Takeaways

Blockchain Basics — Remember

- Blocks link by hash: each commits to its predecessor, making history tamper-evident.
- Merkle trees compress thousands of transactions into one root with log(n) proofs.
- Transactions are signed intents; the mempool auctions block space by fee.
- PoW spends energy, PoS risks capital — both buy Sybil resistance for consensus.
- Finality is probabilistic on PoW (deeper = safer) and absolute under BFT voting.
- Full nodes verify everything; light clients verify proofs; gossip spreads it all.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev